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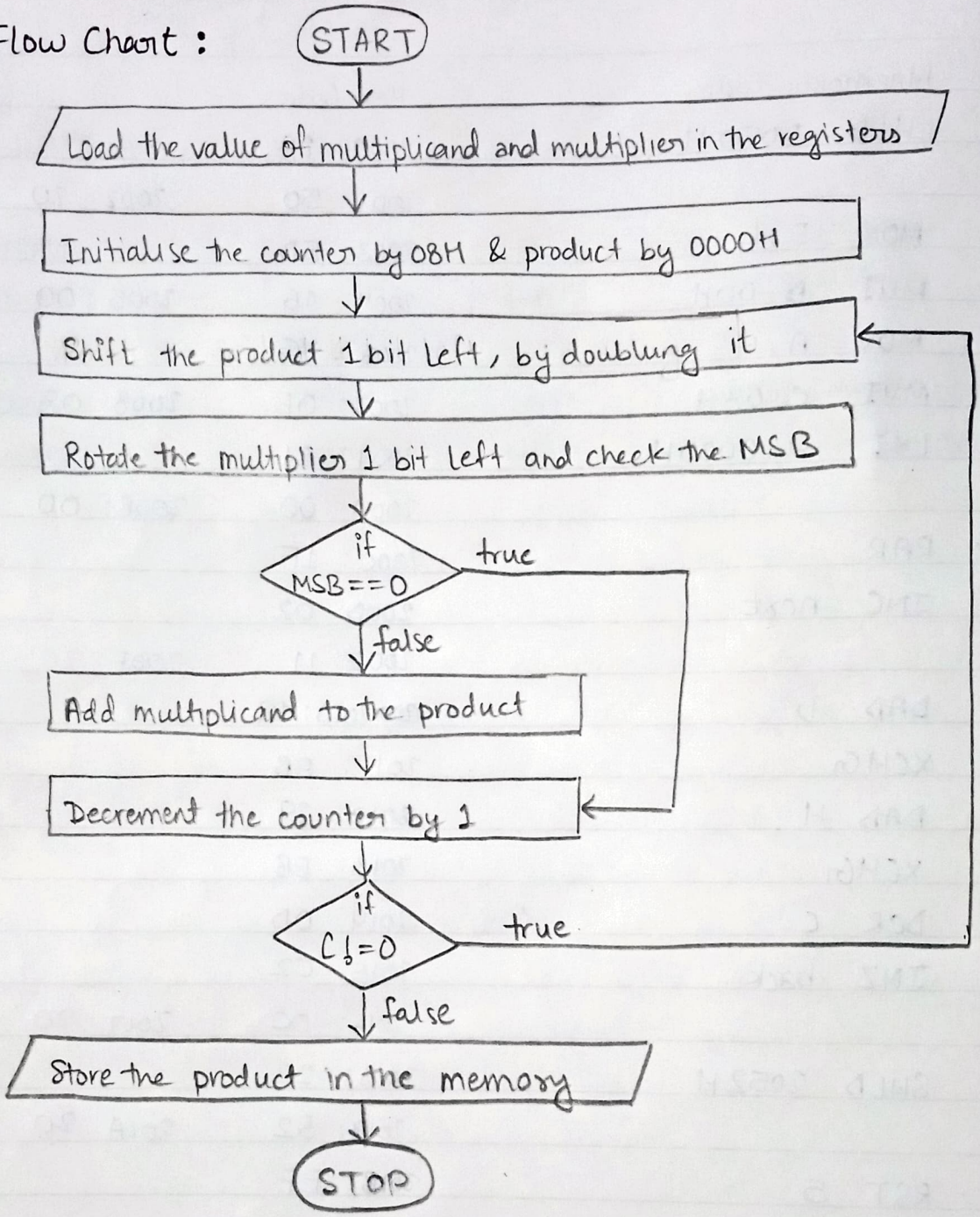
- Aim: Write a program in assembly language of 8085 to find the 16-bit product of two 8 bit numbers by shifting the product.

- Registers Used: Accumulator : Multiplier
  E reg : Multiplicand
  C reg : Counter
  HL pair : Product

- Algorithm: (i) start and initiate the address of the program
(ii) Load the content from mem. addr. 2051 & 2050 to H&L registers resp.
(iii) move the value of L to E reg, and initialise D reg with 00H
(iv) move the value of H reg to Accumulator, and initialise C reg with 08H.
(vi) Initialise the HL reg pair with value 0000H.
(vii) shift the bits of the product stored in HL reg pair, by doubling it.
(viii) Obtain the left most bit of multiplier stored in A into carry flag.
(ix) If MSB is 1, add the multiplicand to the product stored in HL pair.
(x) decrement C reg. by 1, if C is not zero, jump to step (vii)
(xi) store the value of H&L registers (i.e. upper & lower bytes of the product) to mem. addr 2053H & 2052H resp.
(xii) Stop.

Teacher's Signature : _____

- Flow Chart :


START
Load the value of multiplicand and multiplier in the registers
Initialise the counter by 08H & product by 0000H
Shift the product 1 bit Left, by doublung it
Rotate the multiplier 1 bit left and check the MSB
if MSB==0
true
false
Add multiplicand to the product
Decrement the counter by 1
if C↓=0
true
false
Store the product in the memory
STOP

• Observation Table

Memory Addr	Before	After	Decimal equv.
2050	C8	C8	C8 ⟶ 200
2051	64	64	64 ⟶ 100
2052	00	20	4E20 ⟶ 20000
2053	00	4E	(∵ 200×100 = 20000)

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- Code:

Mnemonic Code	Hex Code	
LHLD 2050H	2000 : 2A	
	2001 : 50	2002 : 20
MOV E,L	2003 : 5D	
MVI D,00H	2004 : 16	2005 : 00
MOV A,H	2006 : 7C	
MVI C,08H	2007 : 0E	2008 : 08
LXI H,0000H	2009 : 21	
	200A : 00	200B : 00
back : DAD H	200C : 29	
RAL	200D : 17	
JNC next	200E : D2	
	200F : 12	2010 : 20
DAD D	2011 : 19	
next : DCR C	2012 : 0D	
JNZ back	2013 : C2	
	2014 : 0C	2015 : 20
SHLD 2052H	2016 : 22	
	2017 : 52	2018 : 20
RST 5	2019 : EF	

- Result : This program is working good for finding the 16-bit product of two 8 bit numbers (without shifting multiplicand)

Teacher's Signature : _____